Conditional Probability	$P[A \mid B] = \dfrac{P[A \cap B]}{P[B]}$	(1)
Multiplication Rule	$P[A \cap B] = P[A \mid B]\ P[B]$	(2)
Bayes Theorem	$P[B \mid A] = \dfrac{P[A \mid B]\ P[B]}{P[A]}$	(3)
Law of Total probability	$P[B] = P[B \mid A]\ P[A] + P[B \mid A^c]\ P[A^c]$	(4)
	$P[B] = P[B \cap A] + P[B \cap A^c]$	(5)
Independence	$P[A \mid B] = P[A]$	(6)
	$P[A \cap B] = P[A]\ P[B]$	(7)

It is known that P(E) = 0.6.
What can we say about P[E|F] when E and F are mutually exclusive?

$$P[E \mid F] = \frac{P(E \cap F)}{P[F]}$$

Mutually exclusive:

$E \cap F = \phi$ (empty intersection)

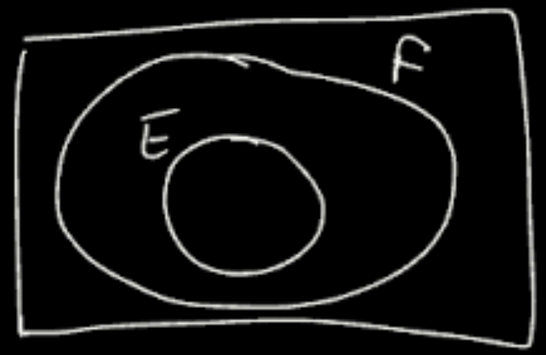
$P[E \cap F] = 0$

$$P[E \mid F] = \frac{0}{P[F]} = 0$$

Conditional Probability	$P[A \mid B] = \dfrac{P[A \cap B]}{P[B]}$
Multiplication Rule	$P[A \cap B] = P[A \mid B]\ P[B]$
Bayes Theorem	$P[B \mid A] = \dfrac{P[A \mid B]\ P[B]}{P[A]}$
Law of Total probability	$P[B] = P[B \mid A]\ P[A] + P[B \mid A^c]\ P[A^c]$
	$P[B] = P[B \cap A] + P[B \cap A^c]$
Independence	$P[A \mid B] = P[A]$
	$P[A \cap B] = P[A]\ P[B]$

It is known that P(E) = 0.6.
What can we say about P[E|F] when E is a subset of F?

$$P(E \mid F) = \frac{P[E \cap F]}{P(F)} = \frac{P(E)}{P(F)}$$



$$= \frac{0.6}{P(F)} \geq 0.6$$

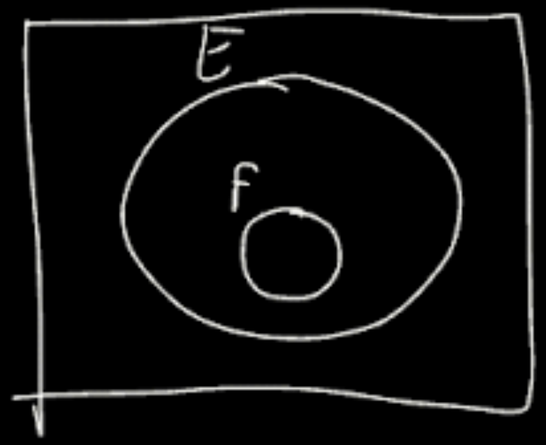
$$E \cap F = E$$

Conditional Probability	$P[A \mid B] = \dfrac{P[A \cap B]}{P[B]}$
Multiplication Rule	$P[A \cap B] = P[A \mid B]\ P[B]$
Bayes Theorem	$P[B \mid A] = \dfrac{P[A \mid B]\ P[B]}{P[A]}$
Law of Total probability	$P[B] = P[B \mid A]\ P[A] + P[B \mid A^c]\ P[A^c]$
	$P[B] = P[B \cap A] + P[B \cap A^c]$
Independence	$P[A \mid B] = P[A]$
	$P[A \cap B] = P[A]\ P[B]$

Let P(E) = 0.6

What can we say about P(E|F) when F is a subset of E

$$P[E \mid F] = \frac{P[E \cap F]}{P[F]} = \frac{P[F]}{P[F]} = 1$$



$$E \cap F = F$$

M and N are two independent events such that P(M U N) = 0.4 and P(M) = 0.3. What can be the value of P(N)?

Conditional Probability	$P[A \mid B] = \dfrac{P[A \cap B]}{P[B]}$
Multiplication Rule	$P[A \cap B] = P[A \mid B]\ P[B]$
Bayes Theorem	$P[B \mid A] = \dfrac{P[A \mid B]\ P[B]}{P[A]}$
Law of Total probability	$P[B] = P[B \mid A]\ P[A] + P[B \mid A^c]\ P[A^c]$
	$P[B] = P[B \cap A] + P[B \cap A^c]$
Independence	$P[A \mid B] = P[A]$
	$P[A \cap B] = P[A]\ P[B]$

$$P(M \cap N) = P(M)\ P(N) = (0.3)\,P(N)$$

$$P(M \cup N) = P(M) + P(N) - P(M \cap N)$$

$$0.4 = 0.3 + P(N) - (0.3)\,P(N)$$

$$0.1 = (1 - 0.3)\,P(N)$$

$$\frac{1}{7} = P(N)$$

The probabilities that "A" and "B" will tell the truth are 2/3 and 4/5 respectively. While acting as a witness in the court what is the probability that

i) they agree with each other

ii) they contradict each other.

Sample space : $\{TT, TF, FT, FF\}$

$P(TT) = \left(\frac{2}{3}\right)\left(\frac{4}{5}\right)$

$P(FF) = \left(1 - \frac{2}{3}\right)\left(1 - \frac{4}{5}\right)$

Agree $= P(TT) + P(FF)$

Recap: Binom:

2 outcomes Eg: H & T
T & F
Pass & Fail
Red & Blue
Success & Fail

Bernoulli trial
do this once

Repeatedly do this experiment "n" times

Eg: 10% chance of success in interview
3 interviews ($n=3$)

parameters
$p = 0.1$
$n = 3$

Possible no of offer letters? $\{0, 1, 2, 3\}$

Sample space = $\{SSS, SSF, SFS, SFF, FSS, FSF, FFS, FFF\}$

$X = 0 \rightarrow \{FFF\}$ 1 way ${}^3C_0 (0.9)^3$

$X = 1 \rightarrow \{FSF, FFS, SFF\}$ 3 way ${}^3C_1 (0.1)(0.9)$

$X = 2 \rightarrow \{SSF, SFS, FSS\}$ 3 ways ${}^3C_2 (0.1)^2 (0.$

$X = 3 \rightarrow \{SSS\}$ 1 way ${}^3C_3 (0.1)^3$

$\text{binom.pmf}\begin{pmatrix} n=3 \\ k=0 \\ p=0.1 \end{pmatrix}$ $\text{binom.pmf}\begin{pmatrix} n=3 \\ k=1 \\ p=0.1 \end{pmatrix}$

$$P[X \leq 2] = P[X=0] + P[X=1] + P[X=2]$$

$$P[X \geq 2] = 1 - P[X \leq 1]$$

Baby health

→ Height & Weight

$\downarrow$ μ_H avg height

$\downarrow$ μ_W avg weight

$z_H = \frac{h - \mu_H}{\sigma_H}$ σ_H std dev

σ_W std weight

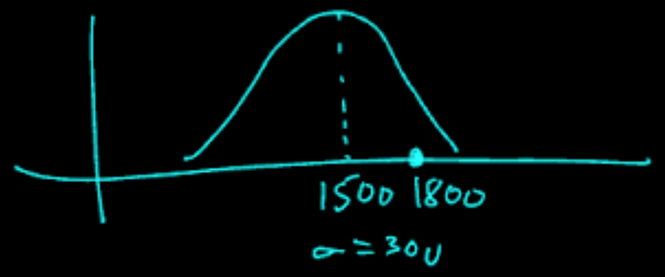
$z_W = \frac{w - \mu_W}{\sigma_W}$

underweight
normal
overweight / Chubby
tall & underw
big

z_H	z_W	
0	0	normal
1.5	0	tall, underweight
0	2	chubby
1	1	big
−1	−1	small
1	−1	underweight

Two types Exam : diff range

	Test 1	Test 2
mean	1500	21
std dev	300	5



Mr A. 1800 on Test 1 $\quad z_A = \frac{1800 - 1500}{300} = \frac{300}{300} = 1$

Mr B: 24 on Test 2 $\quad z_B = \frac{24 - 21}{5} = \frac{3}{5}$

Who is better?